

Smart Meters in London data files: roles, importance, and what Section 4 shows

Sep 28, 2026 · @Boss

Overview

Our data comes from two sources: the Kaggle dataset "Smart Meters in London" (version 21, 343 files, 9.56 GB, almost all CSV) and the dToU price schedule (`Tariffs.xlsx`) from the London Datastore. The schedule must be downloaded separately because Kaggle does not include it.

The repo `purin-kan/BigDataProj` holds no CSV files, because the data is far too large to commit to GitHub. The notebook `src/01_eda_smart_meters.ipynb` uses kagglehub to download it into a local cache, so each teammate needs their own Kaggle API token to run it.

CSV is only the raw format. The pipeline uploads the CSVs to S3 and converts them once to Parquet, partitioned by year and month. Every later step reads Parquet.

Our question is how much the 1,123 dToU households cut or shifted consumption during High and Low price events, compared with the 4,443 Std households in the same half-hours. Answering it takes three kinds of data:

1. **How much** — each household's consumption in each half-hour
2. **Who** — which households were on dToU and which on Std
3. **When** — which half-hours of 2013 were High, Normal or Low

We also need controls (weather, holidays) so that environmental effects are not counted as price effects.

What each file does and how much it matters

Three files are essential: `halfhourly_dataset/` (how much), `informations_households.csv` (who) and `Tariffs.xlsx` (when). The rest support the analysis or help interpret results.

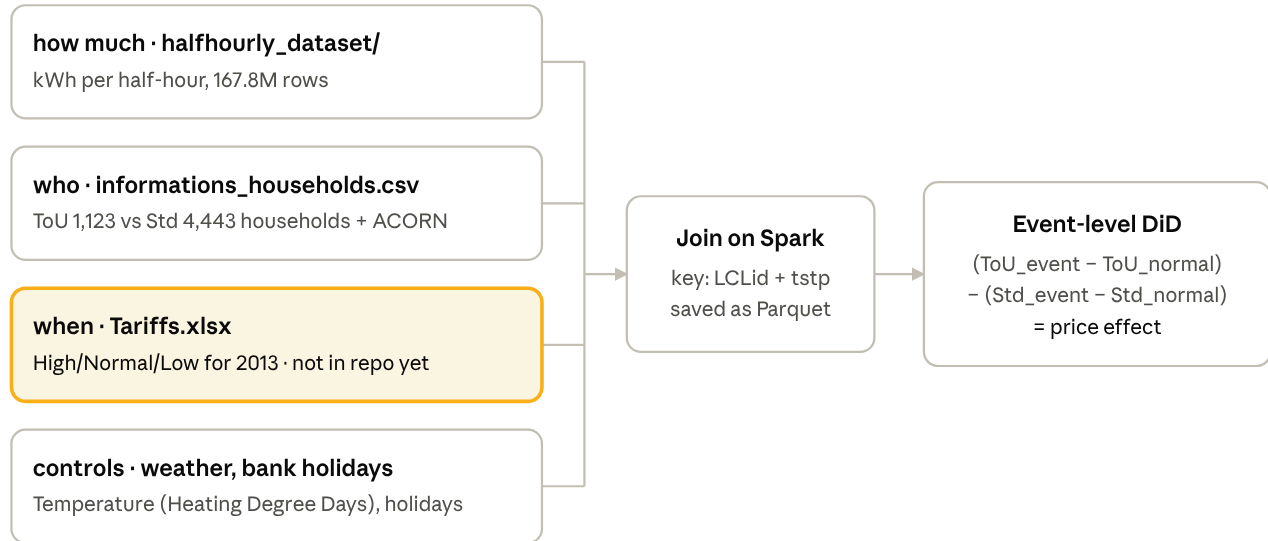
File	Contents	Role in our study	Importance
<code>halfhourly_dataset/block_*.csv</code> (112 files)	167,817,021 readings: <code>LCLid</code> , <code>tstp</code> , <code>energy(kWh/hh)</code>	The outcome. Price windows are set in half-hours, so only this file is fine-grained enough to show load cut or shifted inside a window.	Essential
<code>informations_households.csv</code>	5,566 rows: <code>LCLid</code> , <code>stdorToU</code> , <code>Acorn</code> , <code>Acorn_grouped</code> , <code>file</code>	Splits households into treatment (ToU, 1,123) and control (Std, 4,443), supplies ACORN for controls and subgroups, and maps each household to its block file. Only 0.2 MB.	Essential
<code>Tariffs.xlsx</code> (London Datastore)	Which half-hours of 2013 were High, Normal or Low	Labels the price windows. Without it the primary event-level DiD cannot be computed. Not in Kaggle, and no code in the repo loads it yet.	Essential

File	Contents	Role in our study	Importance
hhblock_dataset/block_*.csv (112 files)	Same readings, one row per household-day with 48 columns hh_0 – hh_47	Ready-made daily load profiles for K- Means and peak-share features.	Convenient
daily_dataset/ (112 files)	Daily sum, mean, median, max, min, std and energy_count per household	Coverage checks (energy_count should be 48) and quick EDA. Too coarse to see a price window.	Supporting
daily_dataset.csv	Same data as daily_dataset/ in one file	Not needed; it is a duplicate.	Not needed
weather_hourly_darksky.csv , weather_daily_darksky.csv	Hourly and daily temperature, humidity, wind and more	Controls; used to compute Heating Degree Days.	Supporting
uk_bank_holidays.csv	25 holiday dates	Control (holiday flag).	Supporting
acorn_details.csv	826 attribute indices per ACORN category (UK average = 100)	Interpretation only. Values are per category, not per household, so joining it gives a model nothing beyond the Acorn label.	Reference

How the files solve our problem

Our primary method, the event-level DiD, needs all three core files together; without any one of them it cannot be computed.

Price effects need all three core files, but Tariffs.xlsx is missing



data files → join → event-level DiD

Std households lived in the same city, saw the same weather on the same dates, and paid a flat price throughout. If we take the ToU gap between event days and Normal days and subtract the same gap for Std households, what remains is the effect of price alone.

Each file fills a different part of that equation:

- `informations_households.csv` says which side of the equation each household is on (ToU or Std)
- `halfhourly_dataset/` supplies each household's consumption in each half-hour
- `Tariffs.xlsx` says which half-hours are events (High/Low) and which are Normal
- Weather and holiday files control for the remaining factors in the panel regression

The gap: `Tariffs.xlsx` is not in Kaggle; it comes from the [London Datastore](#). ToU households received day-ahead signals of High (67.20p), Normal (11.76p) or Low (3.99p) per kWh, while Std households paid a flat 14.228p per kWh.

No code in the repo loads this file yet. Until it is merged, these parts of Section 5 cannot run:

- the event-level comparison (primary method)
- EDA split by High/Normal/Low days
- rebound analysis around events
- responsive/non-responsive labels for classification

What remains is the year-over-year DiD, which per Section 4.4 only 489 Std and 86 ToU households qualify for.

Join caveat: check that the price schedule and the `tstp` column use the same clock (GMT or British Summer Time). A one-hour offset would misplace every summer price window.

Section 4: which file each table and figure uses, and what it shows

Section 4 touches every file, but the numbers that drive Section 5 decisions come from four: `halfhourly_dataset/`, `informations_households.csv`, `daily_dataset/` and `hhblock_dataset/`. The weather, holiday and `acorn_details.csv` files appear only in row counts, data types and missing-value checks.

Table/figure	Files used	What it shows	What it tells us
Table 4.1 (4.1)	All files	Rows and size per file: 343 files, 9.56 GiB	<code>halfhourly_dataset/</code> holds 96% of rows and 77% of bytes, so the heavy work sits in one layer and needs Spark. The daily and hhblock layers summarise or reshape the same readings, so they are left out of totals.
Figure 4.1 (4.1)	All files except the duplicate <code>daily_dataset.csv</code>	Log-scale bar chart of records per file	Sizes range from 25 to 167.8 million rows; the small household and weather files load fine in pandas.

Table/figure	Files used	What it shows	What it tells us
Table 4.2, first table (4.2)	<code>halfhourly_dataset/</code> , <code>informations_households.csv</code>	Columns, types and value ranges of the two core files	<code>energy(kWh/hh)</code> loads as text because missing values are the string <code>"Null"</code> , fixed with <code>na_values=["Null"]</code> ; <code>tstp</code> must be parsed to datetime before any join.
Table 4.2, second table (4.2)	<code>daily</code> , <code>hhblock</code> , <code>weather</code> , <code>holidays</code> , <code>acorn_details.csv</code> , <code>dToU schedule</code>	Columns of the supplementary files we use	Shows which files are controls, which are reference only, and that the dToU schedule comes from the London Datastore.
Table 4.3 (4.3)	Every file with missing values	Missing counts per column and how we handle them	Only 5,560 <code>"Null"</code> values (0.003%), but 683,763 whole rows are absent (0.41%), so cleaning must detect missing timestamps, not just nulls. The 51 households without ACORN are excluded from group comparisons.
Figure 4.2 (4.4)	Per-household consumption + <code>stdorToU</code>	Households reporting per month, Std vs ToU	Most households joined during 2012, so few have a full 2012 baseline: only 489 Std and 86 ToU qualify for both years, against 4,071 Std and 1,038 ToU for 2013. This is why the event-level DiD is the primary method.

Table/figure	Files used	What it shows	What it tells us
Figure 4.3 (4.4)	<code>daily_dataset/</code> (<code>energy_count</code>)	Share of household-days with fewer than 48 readings, per month	22% incomplete in Nov 2011, about 1% or less through 2013, so the trial-year data is clean. hhblock is exactly 41,081 rows shorter than daily, so it holds complete days only.
Figure 4.4 (4.5)	<code>informations_households.csv</code>	Std 79.8% vs ToU 20.2%	The control group is about four times the size of the treatment group.
Figure 4.5 (4.5)	<code>informations_households.csv</code>	Households per ACORN category, coloured by group	ACORN-E (1,567) and ACORN-Q (831) are the largest; 51 households are unclassified.
Figure 4.6 (4.5)	<code>informations_households.csv</code>	dToU share per ACORN group, with a χ^2 test	Affluent 22.4%, Comfortable 21.4%, Adversity 16.4% ($p = 6.3 \times 10^{-6}$). Assignment is unbalanced, so Section 5 controls for ACORN group.
Figure 4.7 (4.5)	<code>hhblock_dataset/</code>	Mean consumption per half-hour across the day	Low of 0.113 kWh at 04:00, peak of 0.321 kWh at 19:00–19:30 (2.8×). The real peak runs 18:00–21:30, but our fixed 16:00–19:00 window covers only 2 of those 7 half-hours.

Table/figure	Files used	What it shows	What it tells us
Table 4.5 (4.5)	<code>daily_dataset/</code> (<code>energy_sum</code>), <code>halfhourly_dataset/</code>	Summary statistics for daily and half-hourly consumption	Right-skewed: mean 10.12 kWh/day vs median 7.82; the largest day is 332.6 kWh, over 40× the median, so outliers need review before modelling.
Figure 4.8 (4.5)	<code>halfhourly_dataset/</code> (10% sample)	Histogram of half-hourly readings (log scale)	Confirms the same right skew at half-hour level: mean 0.209 kWh is 1.8× the median of 0.113 kWh.

Fixes needed in the Section 4 write-up

Three issues a marker will see in the report:

- ☐ The supplementary-files table in 4.2 is labelled "Table 4.2", duplicating the first table. The 4.2 intro already calls it Table 4.3, so only the label needs changing. Then renumber the Missing Values table to Table 4.4 and change "Table 4.3 lists every column..." in 4.3 to 4.4.
- ☐ Table 4.5 still has `[Q1]` , `[MEDIAN]` and `[Q3]` placeholders in the half-hourly column, although the text below it already gives the sample median (0.113 kWh). Fill in the median and compute Q1/Q3 from the same 10% sample.
- ☐ The text refers to Figure 4.8, but the figure is not in the report yet. Generate and insert the histogram before submitting.